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College of Science
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Physics 180 THV-1
Problem Set 10

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Problem 10.1

Show that the general solutions of the free-particle Dirac equation for a particle with momentum $\mathbf{p}$ are obtainable from the solutions for a particle at rest via Lorentz transformations. Hint: What are the Lorentz transformation properties of the Dirac spinors?

At rest:

$$u_1(m, 0) = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad u_2(m, 0) = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad u_3(m, 0) = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad u_4(m, 0) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (1.1)$$

Moving (i.e., there is a boost, like *riding a rocket*):

$$u_1 = N \begin{bmatrix} 1 \\ 0 \\ \frac{p_z}{E+m} \\ \frac{p_x + ip_y}{E+m} \end{bmatrix}, \quad u_2 = N \begin{bmatrix} 0 \\ 1 \\ \frac{p_x - ip_y}{E+m} \\ -\frac{p_z}{E+m} \end{bmatrix}, \quad u_3 = N \begin{bmatrix} \frac{p_z}{E+m} \\ \frac{p_x + ip_y}{E+m} \\ 1 \\ 0 \end{bmatrix}, \quad u_4 = N \begin{bmatrix} \frac{p_x - ip_y}{E+m} \\ -\frac{p_z}{E+m} \\ 0 \\ 1 \end{bmatrix} \quad (1.2)$$

Solution

Proof. As hinted, a key point is that the Dirac spinors transform under the spinor representation of the Lorentz group. A boosted solution is therefore obtained by applying the corresponding spinor boost to a rest-frame solution.

Consider the four-momentum of a particle at rest, p_0^μ :

$$p_0^\mu = (m, \mathbf{0}) \quad (1.3)$$

Boosting it would lead to a new momentum and energy

$$p^\mu = (E, \mathbf{p}) \quad (1.4)$$

where $E = \sqrt{\mathbf{p}^2 + m^2}$

The spinor transformation is then

$$u(p) = S(\Lambda)u(m, 0) \quad (1.5)$$

Where $S(\Lambda)$ comes from how a Dirac spinor transforms

$$\psi' = S(\Lambda)\psi$$

and has values

$$S(\Lambda) = \exp\left(-\frac{i}{4}\omega_{\mu\nu}\sigma^{\mu\nu}\right)$$

where $\omega_{\mu\nu}$ are the six Lorentz parameters, and the spinor generators are:

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$$

For a pure boost, the Lorentz parameters are $\omega_{0i} = \phi \hat{n}_i$ with all $\omega_{ij} = 0$, so

$$S(\Lambda) = \exp\left(-\frac{i\phi}{2} \hat{n}_i \sigma^{0i}\right).$$

Using $\sigma^{0i} = i\gamma^0\gamma^i = i\alpha^i$, where $\alpha^i \equiv \gamma^0\gamma^i$, this becomes

$$S(\Lambda) = \exp\left(\frac{\phi}{2} \hat{\mathbf{n}} \cdot \boldsymbol{\alpha}\right).$$

Since $(\hat{\mathbf{n}} \cdot \boldsymbol{\alpha})^2 = I$, the exponential linearizes:

$$S(\Lambda) = \cosh\left(\frac{\phi}{2}\right)I + \sinh\left(\frac{\phi}{2}\right)(\hat{\mathbf{n}} \cdot \boldsymbol{\alpha}).$$

At this stage, the transformation is specified by the boost direction $\hat{n}$ and rapidity ϕ . We now trade these for the components of the four-momentum $\mathbf{p}$. The boost takes the rest momentum $p_0^\mu = (m, \mathbf{0})$ to $p^\mu = (E, \mathbf{p})$, so the rapidity satisfies

$$\cosh \phi = \frac{E}{m}, \quad \sinh \phi = \frac{|\mathbf{p}|}{m},$$

and the boost direction is $\hat{n} = \mathbf{p}/|\mathbf{p}|$. The half-angle identities then give

$$\cosh\left(\frac{\phi}{2}\right) = \sqrt{\frac{E+m}{2m}}, \quad \sinh\left(\frac{\phi}{2}\right) = \frac{|\mathbf{p}|}{\sqrt{2m(E+m)}}.$$

Substituting and factoring out $\sqrt{(E+m)/2m}$, we obtain the spinor boost:

$$S(\Lambda) = \sqrt{\frac{E+m}{2m}} \left(I + \frac{\boldsymbol{\alpha} \cdot \mathbf{p}}{E+m} \right).$$

Since a given four-momentum $p^\mu = (E, \mathbf{p})$ with $p^\mu p_\mu = m^2$ is reached from the rest momentum by a *unique* pure boost Λ_p , the transformation is now fully determined by $\mathbf{p}$ alone. We may therefore replace the argument Λ by the momentum $\mathbf{p}$ and write the spinor boost matrix as

$$S(\mathbf{p}) = \sqrt{\frac{E+m}{2m}} \left(I + \frac{\boldsymbol{\alpha} \cdot \mathbf{p}}{E+m} \right). \quad (1.6)$$

In the dirac representation,

$$\boldsymbol{\alpha} = \begin{bmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{bmatrix},$$

So, we rewrite the spinor boost matrix as

$$S(\mathbf{p}) = \sqrt{\frac{E+m}{2m}} \begin{bmatrix} I & \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} & I \end{bmatrix} \quad (1.7)$$

Since [1],

$$\boldsymbol{\sigma} \cdot \mathbf{p} = \begin{bmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{bmatrix}$$

We expand the entire boost operator as a 4×4 matrix

$$S(\mathbf{p}) = \sqrt{\frac{E+m}{2m}} \begin{bmatrix} 1 & 0 & \frac{p_z}{E+m} & \frac{p_x - ip_y}{E+m} \\ 0 & 1 & \frac{p_x + ip_y}{E+m} & \frac{-p_z}{E+m} \\ \frac{p_z}{E+m} & \frac{p_x - ip_y}{E+m} & 1 & 0 \\ \frac{p_x + ip_y}{E+m} & \frac{-p_z}{E+m} & 0 & 1 \end{bmatrix}$$

Now we apply this spinor boost operator to the four rest spinors.

Take $u_1(m, 0)$. It is

$$u_1(m, 0) = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Applying the spinor boost operator to u_1 , we obtain

$$\begin{aligned} S(\mathbf{p})u_1(m, 0) &= \sqrt{\frac{E+m}{2m}} \begin{bmatrix} 1 & 0 & \frac{p_z}{E+m} & \frac{p_x - ip_y}{E+m} \\ 0 & 1 & \frac{p_x + ip_y}{E+m} & \frac{-p_z}{E+m} \\ \frac{p_z}{E+m} & \frac{p_x - ip_y}{E+m} & 1 & 0 \\ \frac{p_x + ip_y}{E+m} & \frac{-p_z}{E+m} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\ &= \sqrt{\frac{E+m}{2m}} \begin{bmatrix} 1 \\ 0 \\ \frac{p_z}{E+m} \\ \frac{p_x + ip_y}{E+m} \end{bmatrix} \end{aligned}$$

If we set

$$N = \sqrt{\frac{E + m}{2m}}$$

Then we obtain the general solution for a boost of u_1 .

$$u_1 = N \begin{bmatrix} 1 \\ 0 \\ \frac{p_z}{E + m} \\ \frac{p_x + ip_y}{E + m} \end{bmatrix} \quad (1.8)$$

Which is precisely what we wanted to show.

It can be shown trivially¹. that the rest of the spinor basis $u_i, i = 2, 3, 4$ are *unit basis spinors*² can be written as the i -th column of the spinor boost matrix, multiplied by N . Hence, it follows that

$$u_2 = N \begin{bmatrix} 0 \\ 1 \\ \frac{p_x - ip_y}{E + m} \\ \frac{-p_z}{E + m} \end{bmatrix}, \quad u_3 = N \begin{bmatrix} \frac{p_z}{E + m} \\ \frac{p_x + ip_y}{E + m} \\ 1 \\ 0 \end{bmatrix}, \quad u_4 = N \begin{bmatrix} \frac{p_x - ip_y}{E + m} \\ \frac{-p_z}{E + m} \\ 0 \\ 1 \end{bmatrix} \quad (1.9)$$

Thus, we show the general solutions for the free-particle Dirac equation. $\square$

¹Which is nerdspeak for “I am tired of typing this out, and the next answers are obvious”, lmao

²unit basis vector equivalents? Is this what you call it? idk man I’m a relativist, spinors are beyond me. Dunno what those are, but they look like unit basis vectors so yah that’s what I’ll call em I think

References

- [1] M. Flores. Lecture Slides *Physics 180: Nuclear and Particle Physics*. Digital lecture slides, 2026. Available from course website, [UvLé](#).

Appendices

Code

No code was used for this problem set

Thank you Mr. Kong

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